Climate Vulnerable Transport Infrastructure and Agricultural Market Access: Evidence from River Transport Disruptions in Brazil

Group B

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Abstract

Brazil exports more than half of the world's soybeans and 19% of corn. Much of this trade moves through the Arco Norte, a major agricultural export corridor whose navigability collapses in drought. The same climate shock that depresses crop yields also restricts the rivers carrying the harvest to world markets, so the corridor amplifies rather than buffers climate exposure for farmers. Under arbitrage, the farmgate-to-world price gap is the log iceberg trade cost from origin to world market. Combining ANA water-level gauges, ANTAQ vessel cargo records, CONAB farmgate prices, and IMF world prices in a municipality–commodity–month panel (2015–2023), we microfound a shift-share instrument from barge engineering. A vessel's feasible cargo is mechanically bounded by the shallowest navigable point along its route, so route-minimum water levels interacting with fixed boat-specific loading constraints produce plausibly exogenous variation in port throughput. Conditional on vessel, route, and time fixed effects, minimum water level significantly predicts vessel loading at the 1% level. A 30% drought-induced fall in instrumented throughput widens the farmgate-to-world price gap by 2.4 percentage-points at the median municipality, with the effect roughly 7% larger at the 75th percentile of remoteness from the assigned port. Welfare losses concentrate in the June–October dry season, peaking at 33.9% in September 2016 and 24.7% in July 2020. The results provide the first instrumented estimate of this channel on the Arco Norte and reframe trade infrastructure from a mechanism of climate adaptation into a source of climate exposure, with direct inputs for evaluating Brazil's R$1.29B (£190.40 million) waterway-dredging programme.

1 Introduction

Brazil is a leading exporter of soybeans and corn. In 2024, the country exported $43.4B of soybeans, equivalent to 54% of the world's total exports, and $8.41B of corn, 19% of total world exports (OEC, 2024). Agribusiness, in turn, accounts for a 23.2% share of Brazilian GDP (CEPEA, 2024). The Madeira, Tapajós and Amazonas river network carrying this trade is critical to millions of livelihoods. In 2023, the severe drought in the Madeira meant that Porto Velho hit its lowest recorded water level in 56 years (Souza et al., 2024). The following year, it broke that record again, causing the Society of Ports and Waterways of Rondônia (SOPH) to halt all operations for the first time in its history (DatamarNews, 2024).

These drought shocks were not one-off or confined to Brazilian rivers. Wang et al. (2024) show that 78% of global road and rail assets face sharply rising exposure to climate extremes, whilst similar low-water disruptions have already affected the Mississippi (Lombardi et al., 2022) and Rhine (Vinke et al., 2022). Inland waterways, therefore, show a broader mechanism where climate shocks impair transport infrastructure. World Weather Attribution (Clarke et al., 2024) estimates

that anthropogenic climate change made the 2023 Amazon drought approximately thirty times more likely. Yet despite the scale and frequency of these events, we lack empirical estimates of the costs of river-transport disruption for Brazil's agricultural export sector. This paper asks how drought-induced reductions in river transport affect the inland-to-world price gap for Brazilian agricultural producers, and quantifies the partial-equilibrium welfare cost of corridor failure. The estimates are directly aimed at an ongoing policy debate where the Brazilian government has committed R$1.29B (£190m) to waterway dredging and maintenance across 2023–2025, with R$529m (£78m) in 2025 alone (Ministério de Portos e Aeroportos, 2026). The recovered elasticity helps evaluate such spending in Brazil and elsewhere.

This paper connects two literature strands. The trade-and-climate literature highlights transport infrastructure as a tool of climate adaptation, as a functioning network buffers against local climate shocks through spatial arbitrage (Burgess & Donaldson, 2010; Costinot et al., 2016). Meanwhile, the climate-impact literature shows that single hydrological shocks can constrain multiple economic sectors (Zscheischler et al., 2020), transmitting risk across sectors and borders (Challinor et al., 2018) and along agricultural commodity corridors (Adams et al., 2021; Keys et al., 2025). Bringing these strands together refocuses infrastructure as a channel of exposure rather than adaptation. The same drought shock that reduces production yields also reduces the navigability of the corridor that carries them to world markets. Figure 1 shows this pathway.

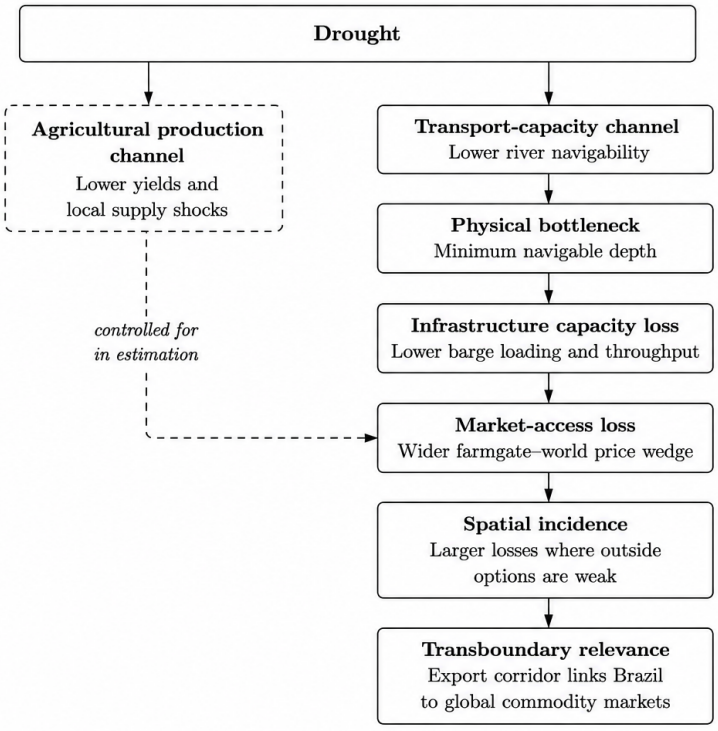


Figure 1: Conceptual transmission pathway from drought to infrastructure-mediated transboundary market-access risk.

The Arco Norte is a clean setting because shocks are large, recurrent, and observable in our data. Figure 2 plots seasonality-adjusted water levels at the gauges on our barge routes. Previous work

treats climate-induced transport disruption as a one-off natural experiment (Steinbach & Zhuang, 2025; Hamano & Vermeulen, 2020; Volpe Martincus & Blyde, 2013), but climate change is making these shocks more frequent and recurrent in expectation. Panel variation across four drought episodes (1) identifies an elasticity embedding producer and shipper responses to recurrent rather than one-off shocks.

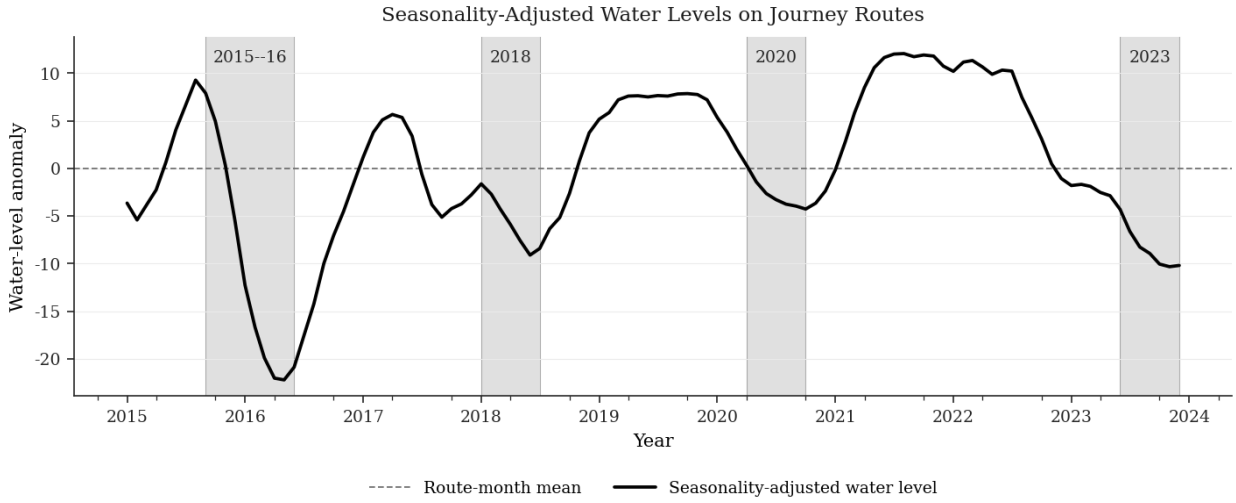


Figure 2: Seasonality-adjusted water levels on barge routes, 2015–2024. The anomaly is the deviation of the route's water level from its long-run average for that calendar month in percentage points. Shaded bands mark the documented low-water episode.

Table 1: Documented Amazon low-water episodes, 2015–2024.

Episode	Basin(s) affected	Timing	Navigation and logistics impact
2015–2016	Madeira, Xingu, Juruá, Purus	broad dry episode	Severe basin-wide drought and transport isolation.
2018	Madeira	dry season	Documented low-water attention levels at Porto Velho and Mamoré on the Madeira mainstem.
2020	Negro, Juruá, Solimões	dry season	Low-water in the western Amazon.
2023	Madeira, Tapajós, Negro, Amazon	broad dry episode	Barges halted, capacity-restricted; Madeira and Tapajós at extreme lows.

Sources: Santos de Lima et al. (2024) document severe Amazon droughts in 2005, 2010, and 2016. The 2018 Madeira episode is documented by Serviço Geológico do Brasil / CPRM (2018) and Governo do Estado de Rondônia (2018). The 2023 navigation restrictions are reported by Mano (2023) and UNICEF (2023). The empirical panel uses only episodes that overlap the price and transport data.

We instrument barge-voyage relative load with the minimum observed water level along each journey and aggregate to a port-level fitted-throughput measure following the exogenous-shifter logic of Borusyak et al. (2021). The instrumented throughput is then related to a municipality-commodity-month panel of log price gaps built from Companhia Nacional de Abastecimento (2024*b*) farmgate prices and International Monetary Fund (2024) world prices.

We make two contributions. Conceptually, climate-vulnerable transport infrastructure transmits

shocks, shifting trade infrastructure from a mechanism of climate adaptation to a source of climate exposure. Empirically, we provide the first instrumented estimate of how climate-induced disruptions affect the inland-to-world price gap, a structural iceberg trade cost (Section 3), through waterway transport on a major agricultural corridor. Existing work examines climate shocks on Brazilian agricultural production; we focus on export-supply-chain vulnerability, finding that a 30% drought-induced reduction in instrumented barge throughput widens the price gap by 2.4 percentage-points at the median municipality, with the effect roughly 7% larger at the 75th percentile of remoteness from the assigned port. Translated into producer revenue, these losses concentrate in the June–October dry-season window. This mechanism is estimated on Brazilian waterways, but its relevance is not limited to this setting. Other agricultural corridors, such as the Mississippi and Rhine (Beuthe et al., 2013; Steinbach & Zhuang, 2025), depend on climate-sensitive inland navigation, whilst global road and rail assets also face increasing exposure (Wang et al., 2024). The estimated elasticity extends to other climate-exposed corridors and opens the door for further work on multi-modal transport vulnerability.

The paper proceeds as follows. Section 2 places the paper within literature. Section 3 develops the theoretical framework. Section 4 describes the data. Section 5 presents the first-stage empirical strategy and results, and Section 6 gives second stage methodology, results, and welfare. Section 8 discusses policy implications, Section 9 reports robustness tests, Section 10 highlights limitations, and Section 11 concludes.

2 Literature Review

2.1 Market access, price gaps, and climate buffering

The paper starts from spatial arbitrage with trade costs. When moving goods across space is costly, infrastructure and geography create transport-cost wedges that reduce trade flows (Limão & Venables, 2001). In agricultural settings, these wedges appear in farm-gate prices: Sotelo (2020) shows that domestic trade frictions generate spatial dispersion in farm-gate prices and shape farmer welfare in Peru. Donaldson (2018) gives this mechanism a welfare interpretation in colonial India, where railroad expansion reduced trade costs, narrowed price gaps, and increased trade. Donaldson & Hornbeck (2016) add the network dimension by showing that infrastructure value depends on market access, not only on whether a place is physically connected.

The trade-and-climate literature uses the same market-access logic in a different direction. In colonial India, railways weakened the link between rainfall failures and famine mortality because grain could move from surplus to deficit regions (Burgess & Donaldson, 2010). This paper studies a different welfare object. We focus on export-crop producer prices rather than subsistence consumption. Spatial arbitrage dampens local price effects of weather shocks, if transport links remain available. River corridors, however, expose a limitation of that buffering logic. Much of the trade-and-climate literature lets productivity vary while bilateral trade costs remain fixed. That separation is less credible when the same drought affects both crops and the transport route carrying them. In the Brazilian Amazon, low water levels affect transport infrastructure at the same time as they affect agricultural output. For export producers, the relevant failure is therefore a market-access failure. The corridor linking interior farm-gate prices to world prices becomes more costly precisely when climate stress is high. The mechanism is consistent with spatial-equilibrium models in which capacity constraints create rents when infrastructure is congested or binding (Fajgelbaum & Schaal, 2020).

2.2 Evidence from low-water and infrastructure disruptions

Mitchell & Biram (2025) are the closest precursor for our setting. Using a difference-in-differences design for Arkansas grain markets, they show that low Mississippi River levels reduce soybean basis most near public river ports. At the −7 foot threshold, the estimated decline is about $0.88 per bushel at 5 miles, falling to $0.44 at 10 miles and $0.18 at 25 miles. This distance gradient indicates that low-water shocks are transmitted through dependence on nearby barge terminals.

That evidence is useful but reduced-form: it does not instrument observed throughput, translate water levels into engineering-predicted capacity, or separate the transport channel from contemporaneous production shocks. Steinbach & Zhuang (2025) provide complementary evidence from the trade-flow side. The 2022 Mississippi drought cut agricultural exports from Louisiana ports by 3.9% between July 2022 and January 2023, with losses peaking at 20.7% in September. The disruption also diverted some shipments, especially wheat, toward East and West Coast ports, while corn and soybeans recovered without persistent rerouting.

Jonkeren et al. (2007) provide the engineering-to-price mechanism formalised in Section 3.4. Rhine low-water episodes reduce vessel load factors and raise per-tonne freight costs. Subsequent Rhine evidence shows that short-run substitution away from barges and replacement capacity are limited (Jonkeren et al., 2011; Schweighofer, 2013). These studies leave open whether hydrology-driven capacity losses widen farmgate-world price gaps.

2.3 Identification and engineering capacity

Building on the market-access and agricultural trade literature (Limão & Venables, 2001; Donaldson, 2018; Donaldson & Hornbeck, 2016; Sotelo, 2020), this paper separates persistent trade-cost geography from temporary transport-capacity shocks. Existing studies show why transport frictions matter for price gaps and welfare, but identify long-run shifts in trade-cost geography from railroads, roads, or network position. We study a different margin which is the short-run failure of an existing export corridor. In river systems, a municipality may remain formally connected to trade even when low water reduces the corridor's effective capacity.

This paper builds on the price-wedge insight but identifies the transport-capacity channel using hydrology-predicted corridor capacity rather than river-stage thresholds alone. Observed shipments are insufficient without exogenous variation. Distance alone is also insufficient because it captures fixed market access rather than month-to-month disruption. The empirical challenge is isolating transport-capacity variation generated by river conditions.

Route-minimum water depth constrains allowable draft, providing the central engineering constraint. In this setting, fixed market access measures exposure to the corridor, while hydrology-predicted capacity provides the time-varying transport shock.

Borusyak et al. (2021) show that a unit-level shift-share IV regression can be represented as an equivalent shock-level regression, so validity can be assessed through the quasi-random assignment of shocks rather than through random assignment of exposure shares. In this setting, fixed market access determines exposure to the export corridor, while hydrology provides the shock to corridor capacity. Here, the shock is physical rather than demand-driven, with hydrological variation affecting the cargo load that a river system can move. Brazil's rich public data environment makes this design feasible, as described in Section 4.

2.4 Brazilian agriculture and transboundary climate risk

Brazil highlights the channel's importance for transboundary climate-risk research. Challinor et al. (2018) argue that climate risk can be transmitted through linked resource systems with global interaction effects. Food-trade assessments apply that logic to commodities. Keys et al. (2025) show that exposure to compound hazards depends on supplier concentration and climate correlation, while Adams et al. (2021) identify Brazil as a major source of those risks in commodity flows. The gap is that these studies mainly trace yield exposure through trade. The transport corridors carrying those flows are also climate-exposed. Wang et al. (2024) show that this is not a river-specific problem as global rail and road infrastructure is projected to face non-linear growth in exposure to record-breaking meteorological hazards emerging as major threats to transport assets. Their analysis stops at exposed infrastructure kilometres rather than estimating the market-access or welfare losses caused when those assets fail. This broadens the empirical need beyond Brazilian rivers as climate shocks increasingly bind physical thresholds across transport modes. Clarke et al. (2024) provide the attribution context for the 2023 Amazon drought. They compare the probability of a drought like the June–November 2023 event in the current climate with its probability in a counterfactual climate without human-induced warming. Their preferred estimates imply that warming made the rainfall-deficit drought about 10 times more likely and the heat-adjusted agricultural drought about 30 times more likely. This motivates the paper by showing rising climate risk in Brazilian agricultural supply chains.

The Arco Norte is central as it is an adaptation investment and a climate-exposed choke-point. The northern routes lower the transportation cost of soybeans and corn from the Center-West to world markets, reducing reliance on longer southern routes (Salin & Somwaru, 2020). That expansion made global market access more dependent on a singular hydrological corridor. Grain-logistics studies emphasise the cost advantage of river-road intermodal chains for Mato Grosso exports routed through the Arco Norte corridor (Branco et al., 2010). However, that advantage depends on navigability. In compound-event terms, Amazonian drought can combine a production shock with a logistics shock. In other words, the same hydrological stress can lower output and shrink river capacity (Zscheischler et al., 2020).

3 Framework: a capacity price-gap elasticity

The framework links droughts to farmgate prices through a transport-capacity channel. When river depth falls, barges load less, effective capacity falls, and the farmgate-to-world price wedge rises. The wedge should rise most for municipalities with weak outside options, motivating a specification in which hydrology-predicted capacity affects price gaps heterogeneously by market access (Figure 1).

We model a partial-equilibrium agricultural export economy in which municipalities indexed by m produce homogeneous traded crops c and ship them to world markets through a single assigned export port $p(m)$. A set of three agents exist in our framework. Producers in municipality m harvest a predetermined quantity $Q^0_{m,c,t}$ in month t and sell at the farm-gate price $P^D_{m,c,t}$. Intermediaries and barge operators move the crop from the farm-gate through the assigned port to the world market, incurring an iceberg trade cost $\tau_{m,c,t} \geq 1$ per unit shipped. World buyers clear the global market at price $P^W_{c,t}$. Under competitive arbitrage along the corridor, the farm-gate price equals the world price divided by the iceberg trade cost:

$$P^D_{m,c,t} = \frac{P^W_{c,t}}{\tau_{m,c,t}}, \qquad \tau_{m,c,t} \geq 1, \tag{1}$$

so the log price gap is

$$\Delta p_{m,c,t} \equiv \ln P^{W}_{c,t} - \ln P^{D}_{m,c,t} = \ln \tau_{m,c,t} \tag{2}$$

Under competitive arbitrage, identification therefore does not require modelling how Brazilian corridor shocks affect world prices (robustness test in Section 9.4).

Equation (1) follows the iceberg trade-cost formulation used in Donaldson (2018). In this setting, the iceberg trade cost is structural object and equivalent to the farmgate-to-world price wedge, that captures all costs between the municipality and the world market.

3.1 Market Access and Low Access

Each municipality is matched to a primary export port $p(m)$. Geographic exposure is measured by inverse distance,

$$\mathrm{MA}^{\mathrm{inv}}_{m} \equiv \frac{1}{d\big(m, p(m)\big)}, \tag{3}$$

where $d(m, p(m))$ is the port-to-municipality distance in kilometres. Higher MA^{inv}_{m} means closer fixed access to the assigned port.

To enter the welfare and second-stage specifications, the inverse-distance measure in equation (3) is transformed into a measure of corridor dependence:

$$\mathrm{LowAccess}_{m} = -\,z\big(\mathrm{MA}^{\mathrm{inv}}_{m}\big), \tag{4}$$

where $z(\cdot)$ denotes standardisation across municipalities in the estimation sample. The sign flip makes higher $\mathrm{LowAccess}_{m}$ correspond to municipalities further from their assigned export port, so the interaction coefficient γ on $\ln \widehat{X}_{m,t} \times \mathrm{LowAccess}_{m}$ reads directly as the additional sensitivity of the price gap to capacity shocks faced by more remote municipalities. $\mathrm{LowAccess}_{m}$ is fixed by geography and historical port-network development, so it does not respond to contemporaneous drought shocks.

3.2 Production as a control for the local supply channel

The same hydrological stress that reduces river navigability also reduces local crop yields through the standard agricultural-production channel, shown in Figure 1. Without controlling for this, β would absorb both responses, and the paper's central claim, that climate shocks raise trade costs through corridor disruption, could not be separated from the conventional production-side story (Costinot et al., 2016). The bottleneck inversion in Section 3.5 makes this explicit where differentiating (5) during a drought delivers two terms with opposite signs, so isolating the capacity channel requires conditioning on the production channel directly.

3.3 Capacity and the price gap

Let $K_{p,t}$ denote effective port capacity in tonnes for the navigable fleet serving port p in month t. We write the log trade cost as a function of capacity, with municipality heterogeneity entering through fixed market access:

$$\ln \tau_{m,c,t} = f\left(\ln K_{p(m),t},\ \mathrm{LowAccess}_{m}\right) + \phi \ln(\mathrm{Prod}_{m,c,t}) + u_{m,c,t}, \tag{5}$$

with sign restrictions

$$\frac{\partial f}{\partial \ln K} < 0, \qquad \frac{\partial^2 f}{\partial \ln K_{p(m),t}\, \partial \text{LowAccess}_m} < 0. \tag{6}$$

The first sign restriction follows from the draft-capacity channel in inland-waterway transport. Jonkeren et al. (2007) show for the Rhine that low water levels reduce vessel load factors and raise freight prices per tonne, while the price per trip changes little. Section 3.4 formalises the same draft-capacity channel. The second restriction captures heterogeneous incidence within the export-corridor hinterland. The market-access measure does not directly observe outside transport options; it ranks municipalities by the fixed cost of reaching the assigned export node. When corridor capacity becomes scarce, lower-cost suppliers are less exposed, while weakly connected interior municipalities become marginal to the corridor. A hydrology-driven capacity loss should therefore be passed back more strongly into farmgate prices where fixed market access is weaker. The sign prediction is $\gamma < 0$: higher predicted capacity narrows the farmgate-to-world price gap more for low-access municipalities, and a drought-induced capacity loss widens it more for the same places.

Unlike Mitchell & Biram (2025), where proximity to a river port proxies barge dependence, distance here captures weak fixed access within a Brazilian export corridor. The sign of the interaction is therefore network-specific: low-access municipalities are more exposed to drought-induced capacity shocks. The function $f(\cdot)$ subsumes congestion, queueing, bargaining, and road-diversion margins in reduced form, the data therefore identifies the total capacity elasticity.

Linearising (5) around the baseline and using (2) gives the local capacity gradient

$$\frac{\partial \Delta p_{m,c,t}}{\partial \ln K_{p(m),t}} = \beta + \gamma\, \text{LowAccess}_m \tag{7}$$

$$\beta \equiv \left.\frac{\partial f}{\partial \ln K_{p(m),t}}\right|_0 < 0 \qquad \text{and} \qquad \gamma \equiv \left.\frac{\partial^2 f}{\partial \ln K_{p(m),t}\, \partial \text{LowAccess}_m}\right|_0 < 0$$

Equation (7) is a first-order approximation of $f(\cdot)$ around the baseline: β and γ characterise the local capacity gradient at average values of $\ln K_{p(m),t}$ and LowAccess_m rather than global properties of the trade-cost function. We return to this assumption in Section 10.3 when assessing how it propagates to the welfare extrapolation. Since $\beta + \gamma\, \text{LowAccess}_m < 0$, a drought-induced fall in capacity ($\Delta \ln K_{p(m),t} < 0$) widens the price gap. As $\gamma < 0$, the widening is largest where LowAccess_m is highest: municipalities farthest from their assigned port, with the weakest access to alternative export options, experience the largest trade-cost increases.

3.4 Engineering source of capacity variation

Capacity varies with hydrology, as a vessel's feasible cargo load is mechanically bounded by the shallowest navigable part of a route. This relationship depends on the vessel's draft, describing the vertical distance from the waterline to the lowest point of the hull. As cargo is loaded, displaced weight increases and the hull sits deeper, so draft rises approximately linearly with cargo through the displacement curve. A boat therefore requires that its loaded draft remains below the available water depth by an under-keel clearance margin (Dorsser et al., 2020).

For barge b on voyage v on date t, let Q_b denote nominal cargo capacity, δ_b^{empty} the empty draft, α_b the load-to-draft conversion (metres per tonne), and u the required clearance. The route-minimum

water level is

$$\theta_{v,t}^{\min} \equiv \min_{g \in R(v)} \theta_{g,t}, \tag{8}$$

where $R(v)$ is the set of gauges along route v. The minimum binds because ample depth elsewhere does not relax the constraint imposed by the single shallowest segment. Solving the safety condition $\delta_b^{empty} + \alpha_b q + u \leq \theta_{v,t}^{\min}$ for maximum realised cargo q and expressing as a fraction of capacity gives the maximum feasible relative load:

$$\mathrm{RL}_{b,v,t}^{max} = \min\left(1,\ \frac{\theta_{v,t}^{min} - \delta_b^{empty} - u}{\alpha_b Q_b}\right) \tag{9}$$

When the bracketed term exceeds one, depth does not bind, and if below one, the maximum load scales with route-minimum depth. This is the first-stage mechanical relevance condition. Identification comes from the interaction of a time-varying route-minimum water-level shock with fixed hull-specific load–draft constraints that map available depth into feasible cargo, rather than from either source alone. Conditional on vessel, route, year, and month fixed effects, the first stage tests whether vessels carry a larger share of capacity when route-minimum water levels relax this mechanical upper bound.

Because fixed voyage costs are spread over tonnes carried, lower load factors raise per-tonne shipping costs and therefore vessel owners aim to maximise the load (Jonkeren et al., 2007). This converts the physical draft inequality (9) into a first-stage relationship between water levels and observed loads:

$$\mathrm{RL}_{b,v,t} = \pi\theta_{v,t}^{min} + \rho \mathrm{Prod}_{p(m),t} + \delta_b + \eta_v + \alpha_{y(t)} + \mu_{m(t)} + \nu_{b,v,t}, \tag{10}$$

where $\mathrm{RL}_{b,v,t}$ is the realised load factor, δ_b is a barge fixed effect, η_v a route fixed effect, $\alpha_{y(t)}$ a year fixed effect, and $\mu_{m(t)}$ is a month fixed effect. Aggregating fitted loads gives hydrology-predicted throughput:

$$\widehat{X}_{p,t} = \sum_{(b,v):\, p(v)=p} Q_b$$

which is the empirical counterpart of port capacity $K_{p,t}$ in the framework.
For municipality m, we define:

$$\widehat{X}_{m,t} \equiv \widehat{X}_{p(m),t}. \tag{12}$$

3.5 The bottleneck inversion

In price-gap notation, a negative production shock raises the local price relative to the world price, so the farmgate-to-world gap narrows:

$$\Delta \ln(\mathrm{Prod}_{m,c,t}) < 0 \quad \text{and} \quad \Delta \ln K_{p(m),t} = 0,$$
$$\Longrightarrow \Delta p_{m,c,t} = \phi\, \Delta \ln(\mathrm{Prod}_{m,c,t}) < 0 \qquad \text{if } \phi > 0$$

Our setting inverts this channel because the same drought also reduces transport capacity. Differentiating (5) gives

$$\Delta p_{m,c,t} = \phi\, \Delta \ln(\mathrm{Prod}_{m,c,t}) + [\beta + \gamma\, \mathrm{LowAccess}_m]\, \Delta \ln K_{p(m),t} + \Delta u_{m,c,t} \tag{13}$$

During a drought, $\Delta \ln K_{p(m),t} < 0$. Since $\beta + \gamma\, \mathrm{LowAccess}_m < 0$, the capacity term is positive: capacity loss widens the price gap. The empirical design isolates the capacity term by using route-minimum water levels to construct hydrology-predicted capacity, $\widehat{X}_{m,t}$, while controlling directly for

local production. This design rules out a simple rainfall instrument: rainfall is not a pure transport shifter when the same Amazonian drought affects both local production and river navigability, activating both terms in (13). Observed throughput is also endogenous because harvest timing and local price conditions affect shipment volumes. We therefore construct hydrology-predicted capacity from route-minimum water levels and relate this measure to municipality-level price gaps. The approach follows exposure designs in which identification comes from plausibly exogenous shocks interacted with fixed exposure or engineering constraints (Borusyak et al., 2021; Goldsmith-Pinkham et al., 2020).

3.6 Mapping to the second stage

Substituting (2) into the linearised capacity equation yields the estimating equation:

$$\Delta p_{m,c,t} = \beta \ln(\widehat{X}_{m,t}) + \gamma \left[\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m\right] + \phi \ln\left(\text{Prod}_{m,c,t}\right) + \alpha_{m,c} + \lambda_{c,t} + \varepsilon_{m,c,t} \tag{14}$$

The municipality-by-crop fixed effect $\alpha_{m,c}$ absorbs permanent route costs. The crop-by-month fixed effect $\lambda_{c,t}$ absorbs world-price movements and crop-specific seasonality. The production control separates the effect of reduced production from the capacity coefficient.

The framework delivers two sign predictions: $\beta < 0$, because higher capacity lowers trade costs and narrows price gaps; and $\gamma < 0$, because low-access municipalities are more exposed to capacity shocks. The iceberg-cost capacity elasticity is:

$$\frac{\partial \Delta p_{m,c,t}}{\partial \ln(\widehat{X}_{m,t})} = \beta + \gamma \, \text{LowAccess}_m \tag{15}$$

With $\beta < 0$ and $\gamma < 0$, throughput losses widen the price gap most for low-access municipalities. In contrast to Donaldson (2018), in which effective distance changes while network capacity remains fixed, our specification utilises drought changes in effective capacity while network geography is fixed.

4 Data

4.1 Data Sources

The analysis combines hydrological, shipments, price, and geospatial datasets with the detail needed to trace the mechanism from river conditions to port capacity and prices. Table 2 provides a summary.

4.1.1 River water levels

Daily water-level readings from ANA HidroWeb cover 2,736 gauge stations across Brazil, giving 5.7 million daily observations (2015-2023) (Agência Nacional de Águas e Saneamento Básico, 2026). Readings are matched to barge journeys for the first-stage regression. These serve as the instrument because hydrology enroute determines the minimum navigable depth along each barge route and thus the maximum feasible cargo load (Jonkeren et al., 2007).

4.1.2 Vessel berthing and cargo records

ANTAQ's administrative registers cover all regulated docking events at Brazilian port facilities (Agência Nacional de Transportes Aquaviários, 2024). The *Atracação* file records 1.28 million berthing events (2010–2025) and *Carga* records commodity classification and tonnes per event, together yielding the voyage-level cargo and capacity records needed for the relative-load outcome. Port coordinates from the ANTAQ registry give origins and destinations for routing, and combine with municipality centroids for the market access measure.

4.1.3 CONAB farmgate prices

Monthly municipality-level farmgate prices from CONAB (Companhia Nacional de Abastecimento, 2024*b*) cover 177,686 observations across 1,013 municipalities and over 100 products (2014–2025). Since export-level prices are not disaggregated below the national level, this is the only publicly available proxy for the domestic side of the inland-to-world price wedge (Margarido et al., 2007), matched to IMF world prices by commodity and month.

4.1.4 IMF world prices and exchange rates

World commodity prices are taken from the IMF Commodity Price System at a monthly frequency (International Monetary Fund, 2024), matched to CONAB prices by commodity and month.

4.1.5 Agricultural controls

Monthly municipality-level EVI from MODIS satellite imagery (Gorelick et al., 2017) serves as the production control, proxying for local agricultural output where monthly administrative production data are unavailable, following Asher & Novosad (2020) who use satellite vegetation indices to measure village-level agricultural yields. Its monthly frequency matches the temporal resolution of the throughput instrument, which annual tonnage-based alternatives cannot provide.

4.1.6 Geographic shapefiles

Waterway shapefiles (Departamento Nacional de Infraestrutura de Transportes, 2024) provide the river network to route vessel journeys and match gauge stations to routes. The IBGE municipality boundary shapefile (Instituto Brasileiro de Geografia e Estatística, 2024) supplies centroids used to compute municipality–port distances and the market access measure.

Table 2: Data Sources

Dataset	Source	Main Content	Frequency	Role
Water levels	ANA HidroWeb	Gauge readings (cm), Coordinates	Daily	Instrument
Vessel berthing	ANTAQ	Berthing events, port code, vessel ID, date	Event-level	First stage
Cargo volumes	ANTAQ Carga	Cargo type, tonnes per voyage	Event-level	First stage
Vegetation index	MODIS via GEE	Monthly EVI by municipality	Monthly	Control
Producer prices	CONAB	Products, municipality, state, date	Monthly	Domestic price
World prices	IMF PCPS	USD/kg prices of products	Monthly	World price
Exchange rates	FRED	Exchange-rate series	Daily	Conversion
Ports file	ANTAQ	Port identifiers, coordinates	Cross-section	Second stage
Municipality centroids	IBGE	Boundary centroids	Cross-section	Market access

Sources: Water levels: Agência Nacional de Águas e Saneamento Básico (2026); Vessel berthing and cargo: Agência Nacional de Transportes Aquaviários (2024); Waterway shapefile: Departamento Nacional de Infraestrutura de Transportes (2024); Vegetation index: Gorelick et al. (2017); Producer prices: Companhia Nacional de Abastecimento (2024*b*); World prices: International Monetary Fund (2024); Exchange rate: Federal Reserve Bank of St. Louis (2024); Municipality centroids: Instituto Brasileiro de Geografia e Estatística (2024).

4.2 Estimation Sample Construction

4.2.1 First-stage journey panel

We construct a journey-level measure of river transport capacity from the berthing, port, and cargo data. For each vessel, we take its maximum observed cargo load as a proxy for maximum capacity. The first-stage journey panel contains 10,342 berthing identifiers across 847 vessels with valid IMO and 47 port facilities, restricted to Brazil–Brazil journeys with valid waterway routes and non-missing water-level percentiles and agricultural production controls. These observations enter the first-stage regression in Table 6. We use this panel to construct the relative load measure following Section 3.6.

$$\text{MaxCapacity}_b = \max_v \text{Cargo}_{b,v} \tag{16}$$

$$\text{RL}_{b,v,t} = \frac{\text{Cargo}_{b,v,t}}{\text{MaxCapacity}_b}, \tag{17}$$

$\text{RL}_{b,v,t}$ captures how intensively a vessel is being used relative to its observed carrying capacity.

We assign each journey a minimum water level reflecting the transport constraint faced by vessels. A routing network is constructed from inland waterways, coastline segments (from Brazil's boundary shapefile), and short connectors between the two (Fajgelbaum & Schaal, 2020). Each origin-destination pair is routed through this network, water-level gauges are matched to the route, and for each journey-date we take the minimum reading across matched gauges. The minimum is the appropriate bottleneck measure as a vessel's feasible load is constrained by the shallowest segment along the route (Jonkeren et al., 2007).

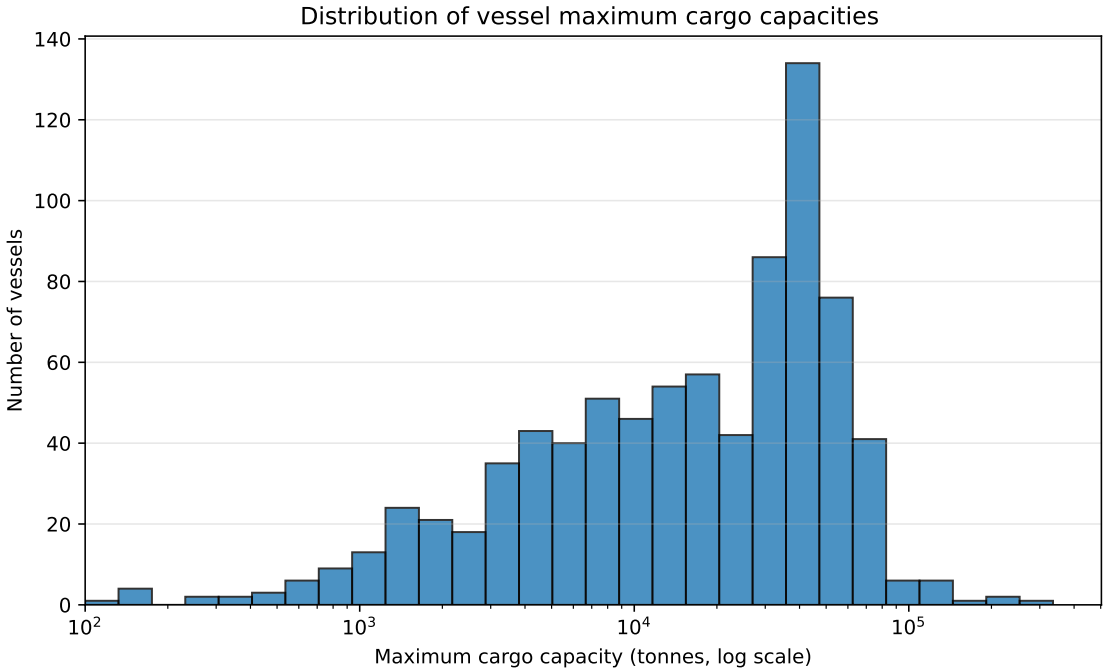


Figure 3: Histogram of Regression Sample Vessels' Max Capacities

The distribution of vessel maximum capacities in Figure 3 spans roughly three orders of magnitude, from small barges of a few hundred tonnes to large ocean-going vessels exceeding 10^5 tonnes. The aggregation in equation (23) weights each journey by its vessel's maximum capacity, so the same percentage-load response to a water-level shock contributes a few hundred tonnes to $\widehat{X}_{p,t}$ for a small barge and several thousand for a fully-rated ocean vessel. MaxCapacity_b is plausibly exogenous to local price-gap shocks because it reflects long-run fleet investment decisions made before any contemporaneous shock, so it acts as a second predetermined source of variation in $\widehat{X}_{p,t}$ alongside the hydrology-driven $\theta^{\min}_{v,t}$.

Table 3: Descriptive statistics

Variable	Mean	SD	p25	Median	p75	N
Relative load (cargo / max)	0.325	0.301	0.056	0.251	0.499	10,342
Min. water level (percentile)	0.406	0.284	0.153	0.375	0.635	10,342
Cargo per journey (tons)	10,787.253	16,680.896	1,024.990	3,733.534	11,717.559	10,342
Vessel max cargo (tons)	37,069.853	55,400.042	7,732.544	23,651.247	40,289.950	10,342
Waterway distance (km)	4,244.901	3,200.078	1,254.420	3,774.205	6,787.925	10,342
Gauges in route buffer	4.444	2.461	2.000	4.000	7.000	10,342

Notes: Sample restricted to BR–BR journeys with valid IMO, successful waterway route and non-missing minimum water-level percentile from 2015-01-01 – 2023-12-31

Table 4: Panel by Year

Year	Journeys	Vessels	Origins	Destinations	Mean MWL	Mean Load
2015	1,514	217	33	32	0.460	0.297
2016	1,188	180	33	33	0.358	0.204
2017	1,003	178	32	32	0.435	0.226
2018	872	178	34	28	0.363	0.301
2019	1,026	185	32	26	0.484	0.300
2020	1,028	208	31	28	0.317	0.300
2021	1,278	237	37	30	0.402	0.439
2022	1,221	204	37	33	0.401	0.405
2023	1,212	240	30	29	0.412	0.423
Total	10,342					

Notes: MWL is the journey-level minimum water-level percentile across gauges in the route buffer. Load is relative load (cargo / vessel max).



Figure 4: Matched Brazilian Water-Level Gauges

4.2.2 Water-level instrument

Of the 2,736 ANA HidroWeb stations, 125 are matched to the inland waterway network within a 10 km buffer, covering 41 river systems. Their daily readings yield 322,383 matched route-level water observations drawn from the wider raw ANA universe of 2,736 gauge stations. Only gauges within the inland-waterway buffer enter the route-level instrument. Figure 4 maps these matched stations against the waterway network.

4.2.3 Second-stage estimation panel

Municipality centroids are matched to the nearest port using haversine distances, and local prices of commodities are matched to world prices. The final second-stage matched port set contains 23 assigned ports. Panel A of Figure 5 maps these matched port facilities. The final sample retains corn and soybean observations matched to first-stage fitted throughput, market access, and production controls, yielding 3,079 municipality–commodity–month observations across 75 municipalities (2015–2023). Table 5 reports summary statistics for the log price gap and market access measure. The municipalities span Brazil's inland agricultural corridor, connected to export markets. Figure 6 maps coverage, showing the sample is concentrated in inland agricultural areas connected to the river-port export network.

Table 5: Summary Statistics: Log price gap and market access

	N	Mean	Std	Min	p25	Median	Max
Panel A: Log price gap ($\ln P^W_{c,t} - \ln P^D_{m,c,t}$)							
All (Corn + Soybeans)	3,079	0.323	1.096	−4.360	−0.323	0.346	3.836
Corn	1,822	0.347	1.094	−4.101	−0.286	0.307	3.604
Soybeans	1,257	0.287	1.099	−4.360	−0.393	0.432	3.836
Panel B: Market access (municipality level, 75 municipalities)							
MA^{inv}_m (raw)	75	0.0068	0.0076	0.0016	0.0029	0.0042	0.0403
LowAccess_m	75	−0.0132	1.0100	−4.4532	−0.0641	0.3330	0.6687

Notes: The sample includes corn and soybeans only (3,079 observations across 75 municipalities from 2015-01 to 2023-12). Panel A reports the raw log price gap $\Delta p_{m,c,t} = \ln P^W_{c,t} - \ln P^D_{m,c,t}$. Panel B reports municipality-level market access: $MA^{inv}_m = 1/d_{m,p(m)}$ is the inverse distance to the assigned port in kilometres; $\text{LowAccess}_m = -z(MA^{inv}_m)$ is the negated z-score so that higher values indicate weaker fixed market access. The z-score standardises within the 75-municipality corn–soybean estimation sample. Two municipalities that appear only in the sorghum panel are excluded.

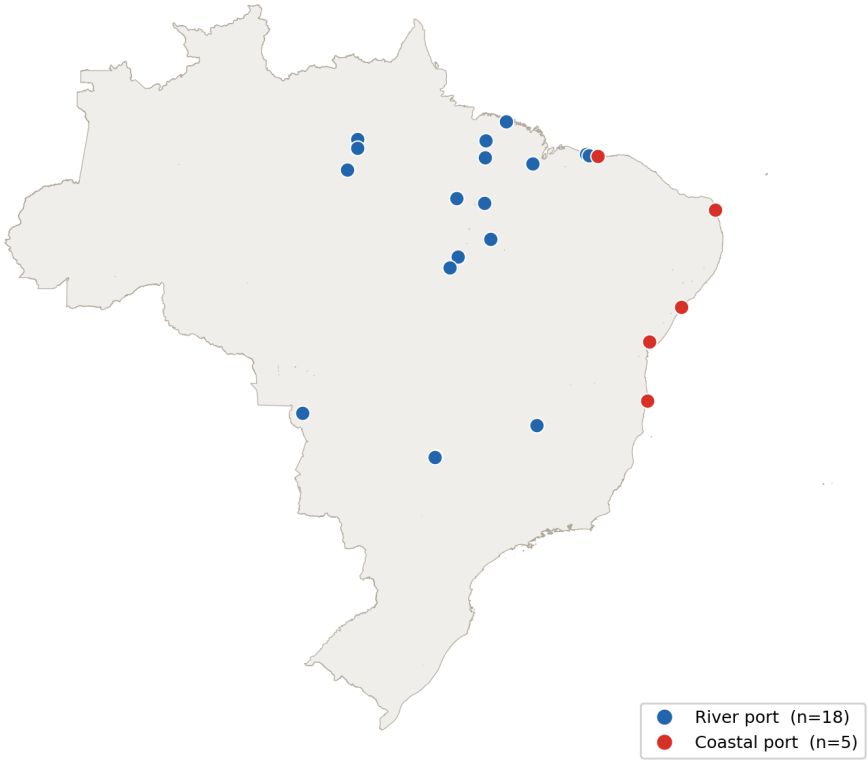


Figure 5: Matched Port Facilities

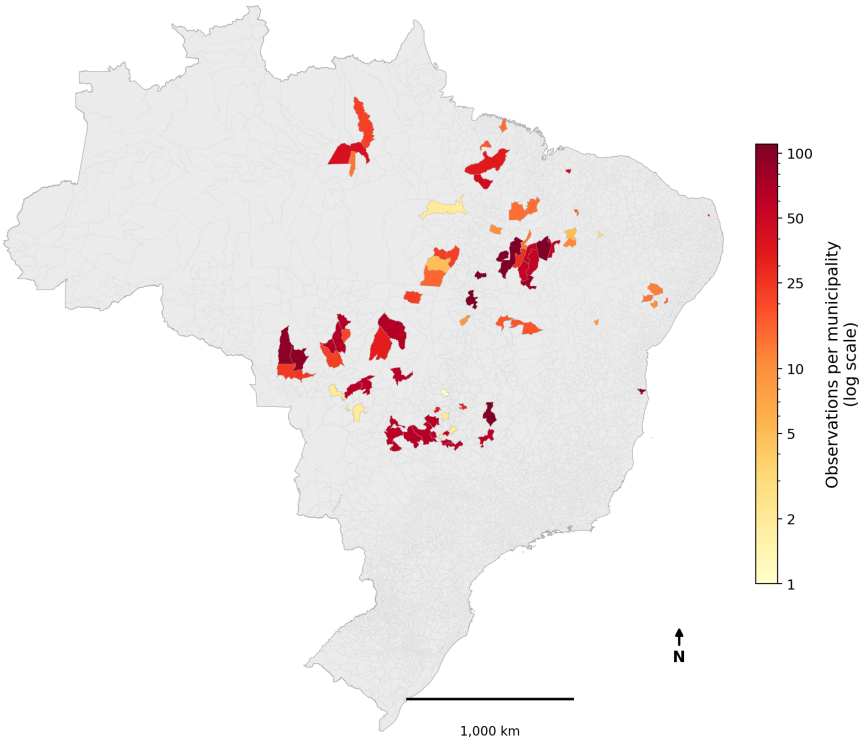


Figure 6: CONAB Corn–Soybean Estimation Municipalities

5 First Stage

The first step in the empirical strategy is to isolate the variation in vessel loading that is attributable to river conditions. We estimate a first-stage regression in which the relative load of each vessel journey is predicted using the minimum water level observed along the route. The resulting fitted values provide an instrumented measure of effective transport capacity, which can then be aggregated and used in the second-stage analysis of the price gap.

5.1 Relevance

Relevance is established empirically by the first-stage regression in Table 6, which shows that minimum water level is a statistically significant predictor of vessel relative load conditional on vessel, route, year, and month fixed effects, with a cluster-robust first-stage F-statistic of 10.53.

The underlying mechanism is physical. A vessel's feasible load is constrained by the shallowest point along its route. When the minimum water level changes, vessels must adjust their load to remain navigable. This engineering link, combined with the loading economics in Section 3.4, motivates the choice of minimum water level along the route as the relevant bottleneck measure. At the port level, aggregating fitted journey-level loads preserves this relevance:

$$\mathbb{E}\left[\theta_{v,t}^{\min} \cdot \widehat{X}_{p,t}\right] \neq 0. \tag{18}$$

5.2 Exogeneity

Exogeneity requires that minimum water level affects the price gap only through its effect on transport capacity, rather than through other channels. The identifying assumption is that river depth at each gauge is determined by enroute hydrology and is plausibly orthogonal to other unobserved determinants of the local price gap, once the second-stage controls and fixed effects are included:

$$\mathbb{E}\left[\theta_{v,t}^{\min} \cdot \varepsilon_{m,c,t} \;\middle|\; \mathbf{Z}_{m,c,t},\, \alpha_{m,c},\, \lambda_{c,m(t)}\right] = 0, \tag{19}$$

where $u_{m,c,t}$ is the second-stage error for municipality m and commodity c in month t, $\mathbf{Z}_{m,c,t}$ collects the agricultural-production and market-access controls, and $\alpha_{m,c}$, $\lambda_{c,m(t)}$ are the municipality–commodity and commodity–month fixed effects of the second stage.

This is supported by the planting-window placebo robustness check in Section 9, which shows that hydrology-driven barge capacity during the planting period has no detectable effect on the contemporaneous price gap ($\widehat{\delta} = 0.005$, $p = 0.76$). This is the direct test of whether the instrument is absorbing unobserved production-side conditions missed by the EVI proxy.

5.3 Specification

The regression estimates whether journeys exposed to higher water levels carry higher relative loads. From Section 3.4, the specification is:

$$\mathrm{RL}_{b,v,t} = \pi\, \theta_{v,t}^{min} + \rho\, \mathrm{Prod}_{p(m),t} + \delta_b + \eta_v + \alpha_{y(t)} + \mu_{m(t)} + \nu_{b,v,t} \tag{20}$$

Vessels differ in age, draft, hull design, ownership, and contracting arrangements, and δ_b absorbs these time-invariant attributes so that π is not contaminated by, for example, larger or newer barges disproportionately operating on deeper-water corridors. Routes differ in natural channel depth, port infrastructure, and typical commodity mix and without η_v, π would partly reflect these cross-sectional differences. Year effects $\alpha_{y(t)}$ absorb common shifts in commodity demand, fleet composition, fuel costs, and macroeconomic conditions. Month effects $\mu_{m(t)}$ remove common seasonal variation in loading patterns.

5.4 Fitted Relative Load

The fitted relative load for journey (b, v, t) is

$$\widehat{\mathrm{RL}}_{b,v,t} = \widehat{\pi}\, \theta_{v,t}^{min} + \widehat{\rho}\, \mathrm{Prod}_{p(m),t} + \widehat{\delta}_b + \widehat{\eta}_v + \widehat{\alpha}_{y(t)} + \widehat{\mu}_{m(t)} \tag{21}$$

Conceptually, $\widehat{\mathrm{RL}}_{b,v,t}$ replaces the realised load factor with the load that hydrology would predict for journey (b, v, t), holding constant the vessel, route, year, and seasonal components.

5.5 Aggregation to port-level

The fitted values are converted to predicted cargo tonnage at the journey level by multiplying by each vessel's maximum cargo capacity,

$$\widehat{\mathrm{Tonnes}}_{b,v,t} = \widehat{\mathrm{RL}}_{b,v,t} \times \mathrm{MaxCapacity}_b, \tag{22}$$

and then summed to the origin-port-by-date level,

$$\widehat{X}_{p,t} = \sum_{(b,v) \in (p,t)} \widehat{\text{Tonnes}}_{b,v,t}. \tag{23}$$

The motivation for this aggregation is the exposure-design framework of Borusyak et al. (2025), in which an aggregate exposure measure is constructed as a fixed-weight sum of underlying shocks, with the aggregate inheriting the identifying variation of its shocks under quasi-random shifter assignment. In our setting, route-minimum water levels $\theta_{v,t}^{min}$ play the role of the underlying shocks, and vessel capacities MaxCapacity_b serve as the fixed weights.

The mapping to Borusyak et al. (2025) is not exact however, because their framework aggregates the shifters themselves, whereas equation (23) aggregates first-stage fitted loads $\widehat{\text{RL}}_{b,v,t}$. Since $\widehat{X}_{p,t}$ is a fixed-weight aggregation of fitted loads driven by route-minimum water levels and predetermined vessel capacities, the aggregation does not introduce an additional source of endogeneity. Conditional on the exogeneity of the underlying water-level shocks and the predetermined nature of the weights, the identifying variation in $\widehat{X}_{p,t}$ inherits the exogeneity of the underlying hydrological variation. The remaining components of $\widehat{\text{RL}}_{b,v,t}$ enter $\widehat{X}_{p,t}$ but are absorbed by the municipality–commodity and commodity–month fixed effects.

5.6 First-Stage Results

Table 6: First-stage regression: Relative load on minimum water level

	Relative load
Minimum water level	0.0216^{***}
	(0.0066)
Agricultural production (origin)	$-4.16\times10^{-6*}$
	(2.42×10^{-6})
Vessel fixed effects	Yes
Route fixed effects	Yes
Year fixed effects	Yes
Month fixed effects	Yes
Observations	10,342
Vessels	847
Routes	177
Time periods (year-months)	108
First-stage F statistic	10.53

Notes: The dependent variable is relative load, defined as total cargo tonnes divided by the maximum cargo tonnes observed for the vessel. Minimum water level is the lowest daily reading, in percentiles, across gauging stations along the journey's waterway route on the journey date. Agricultural production is the EVI value in the municipality nearest to the origin port. The specification includes vessel and route fixed effects, together with year and month dummies (2015 and January omitted). Standard errors clustered by vessel are in parentheses. The first-stage F statistic is the squared cluster-robust t-statistic on the excluded instrument; it clears the conventional Stock–Yogo threshold of 10. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 6 reports the first-stage estimates. The coefficient on minimum water level, statistically significant at the one percent level, is 0.0216. The engineering relationship described above holds, and deeper water along the route allows vessels to carry a higher share of their maximum capacity. Since minimum water level is measured as a within-station percentile rank, the coefficient is interpreted as the change in relative load associated with a one-unit shift in the percentile rank of the binding gauge along the route.

Forty-two percent of vessels operate on a single route, and the average vessel covers 2.7 routes. This pattern motivates the inclusion of route fixed effects alongside vessel fixed effects. While a substantial minority of vessels are route-specialised, making vessel and route effects partly overlap for them, most vessels operate across multiple routes, so the two sets of effects identify different sources of nuisance variation for the bulk of the sample.

6 Second Stage

The second stage implements the estimating equation derived in Section 3.6. We bring the hydrology-predicted port-throughput measure from the first stage to the municipality–commodity–month price panel and estimate whether lower fitted transport capacity is associated with a wider farmgate-to-world price gap. The outcome is the log price gap, while the key regressor is fitted log throughput at the municipality's assigned port. To investigate heterogeneous exposure, fitted throughput is interacted with LowAccess_m, as defined in Section 3.5, so higher values indicate weaker fixed market access.

6.1 Estimation sample

The second-stage regressions use the filtered corn–soybean sample reported in Table 5, retaining only observations matched to fitted port throughput, market access, and production controls.

6.2 Specification

The second stage regresses the log price gap on fitted port throughput from the first stage. It estimates the capacity price-gap elasticity derived in Section 3.6 (equation (14)) on the corn–soybean sample described in Table 5, with standard errors clustered at the municipality level. The pooled specification is:

$$\Delta p_{m,c,t} = \beta \ln\big(\widehat{X}_{m,t}\big) + \gamma\Big[\ln\big(\widehat{X}_{m,t}\big) \times \text{LowAccess}_m\Big] + \phi \ln\big(\text{Prod}_{m,c,t}\big) + \alpha_{m,c} + \lambda_{c,t} + \varepsilon_{m,c,t} \quad (24)$$

6.3 Second Stage Results

6.3.1 Pooled Regression Results

Table 7 reports pooled second-stage estimates of equation(24). The main finding is that lower hydrology-predicted capacity widens the inland-to-world price wedge. Log throughput enters negatively and is significant at 1% across all three columns. The interaction with LowAccess_m is -0.030 at 1% significant level, so municipalities further from their assigned port show larger price-gap responses to the same capacity shock.

Table 7: Second-Stage Results

	Uncentered (1)	Centered at p50 (2)	Centered at p75 (3)	No interaction (4)
$\ln(\widehat{X}_{m,t})$	-0.0571^{***}	-0.0673^{***}	-0.0718^{***}	-0.0577^{**}
	(0.0215)	(0.0220)	(0.0223)	(0.0224)
$\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m$	-0.0295^{***}	-0.0295^{***}	-0.0295^{***}	
	(0.0071)	(0.0071)	(0.0071)	
$\ln(\text{Prod}_{m,c,t})$	-0.1571^{*}	-0.1571^{*}	-0.1571^{*}	-0.1596^{*}
	(0.0865)	(0.0865)	(0.0865)	(0.0861)
Observations	3,079	3,079	3,079	3,079
Municipality × commodity FE	Yes	Yes	Yes	Yes
Commodity × month FE	Yes	Yes	Yes	Yes
Municipalities	75	75	75	75
Municipality commodity entities	115	115	115	115
Clustered SE	Municipality	Municipality	Municipality	Municipality

Notes: The dependent variable is the log municipality-level crop price gap. The sample includes corn and soybeans only (dropping sorghum). LowAccess_m is the negative standardised inverse-distance-to-port measure, so higher values indicate lower fixed market access. Columns (1)–(3) differ only in the centering of LowAccess_m; column (4) omits the interaction term entirely, estimating a common capacity elasticity across all municipalities. Standard errors clustered by municipality are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The three columns report the identical regression with the same fixed effects but evaluated at different points of the market-access distribution. Column (1) evaluates the throughput coefficient at the average municipality. Column (2) centres at the median, and Column (3) is estimated at the 75th percentile of remoteness. Because LowAccess_m is normalised within sampled municipalities, the base coefficient $\widehat{\beta} = -0.057$ in column (1) is the capacity elasticity of the price gap at the average municipality.

The total capacity elasticity follows equation (15):

$$\frac{\partial \Delta p_{m,c,t}}{\partial \ln(\widehat{X}_{m,t})} = \widehat{\beta} + \widehat{\gamma} \times \text{LowAccess}_m$$

This elasticity is structural: by equation (2), it is the elasticity of the log iceberg trade cost $\ln \tau$ with respect to instrumented capacity. At the median municipality, the elasticity is -0.067, and at the 75th percentile of remoteness, it is -0.072. The interaction coefficient $\widehat{\gamma} = -0.030$ implies that each additional standard deviation of distance from port amplifies the throughput effect by 3 percentage-points. A drought event that cuts port loading by 30% hence widens the price-gap by 2.4 percentage-points at the median municipality,[1] while a municipality at the 75th percentile of remoteness faces roughly 7% larger increases. This gap reflects the buffering role of port connectivity in equation (15). A municipality with access to closer and more ports can reroute through alternative corridors, reducing the iceberg trade-cost adjustment when loading capacity falls at the nearest port.

Comparing with theoretical prediction in Section 3, the negative $\widehat{\beta}$ matches the first inequality in

[1] $(\widehat{\beta} + \widehat{\gamma} \times \text{LowAccess}_m) \times \ln(0.7) = -0.067 \times (-0.357) = 0.024$

equation (6), meaning that when hydrology increases the port capacity, the inland-to-world gap narrows, while the negative $\widehat{\gamma}$ matches the cross-derivation in equation (6), meaning the price gap increases most where outside export options are weakest. Our estimation confirms the similar pattern observed in Rhine (Jonkeren et al., 2007), and $\widehat{\beta}$ extends it to price gap their trip-level data cannot reach. This negative interaction coefficient $\widehat{\gamma}$ captures how market access moderates the throughput shocks, consistent with the geographic arbitrage mechanism in Costinot et al. (2016) applied to the transport corridor, but opposite to Mitchell & Biram (2025), whose largest price effects appear on markets closest to Mississippi ports.

The production control $\ln(\text{Prod}_{m,c,t})$ is -0.157, opposite sign to Section 3.5. Because $\ln(\text{Prod}_{m,c,t})$ is included only to control the local production channel, $\widehat{\phi}$ has no causal or structural interpretation and its sign need not match that prediction. The one concern is whether the production proxy is too noisy to capture that channel fully, which would bias β and γ. Section 9.3 addresses this, showing the instrument does not absorb production conditions the control misses.

6.3.2 Commodity-Specific Results

The pooled specification constrains β and γ to be the same across crops. But agriculture products differ in harvest timing and in how much of their output moves by river, so a common elasticity may mask crop-level heterogeneity in the transport channel. Table 8 estimates the specification separately for each commodity with municipality and time fixed effects and clustered standard errors as in the pooled model.

Table 8: Second-Stage Results by Commodity

	Corn	Soybeans
$\ln(\widehat{X}_{m,t})$	-0.0611^{**}	-0.0465^{***}
	(0.0298)	(0.0174)
$\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m$	-0.0319^{***}	-0.0271
	(0.0081)	(0.0238)
$\ln(\text{Prod}_{m,c,t})$	-0.2446^{**}	-0.0269
	(0.1246)	(0.0612)
Observations	1,822	1,257
Municipality FE	Yes	Yes
Time FE	Yes	Yes
Municipalities	70	45
Clustered SE	Municipality	Municipality

Notes: The dependent variable is the log municipality-level crop price gap. Each column estimates the second-stage specification separately by commodity. LowAccess_m is the negative standardised inverse-distance-to-port measure, so higher values indicate lower fixed market access. Standard errors clustered by municipality are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The two products show large differences in the estimates. For corn, $\widehat{\beta}$ is -0.061 at 5% significant level and the market-access interaction is $\widehat{\gamma} = -0.032$. Both signs match the theoretical predictions. The strong interaction term for crop reveals the meaningful impact of market access when normal drought season overlaps with the crop harvest month (February-September) (USDA Foreign Agricultural Service, 2025). During this window, corn producers in remote municipalities face the dual pressure

of peak local supply and shrinking barge capacity, leaving them few alternatives to absorb the cost shock.

For soybeans, the direct effect is $\widehat{\beta} = -0.047$, smaller in magnitude than corn but more precisely estimated, reflecting the larger and more geographically concentrated soybean trade. The interaction term is $\widehat{\gamma} = -0.027$, negative as predicted but not statistically significant. There could be two reasons. First, soybeans are harvested primarily between January and April (USDA Foreign Agricultural Service, 2025), when water levels are still high and river capacity is less constrained, so market access impacts less at the margin. Second, soybean logistics are well-served by alternative road corridors with trucks handling 54% of all Brazilian soybean exports (Péra & Salin, 2024), partially offsetting waterway capacity shocks. We test this road-diversion margin direction in Section 9.2. The same pattern appears in Mitchell & Biram (2025) where Mississippi river terminals with better access to rail show smaller price responses to low water. Even though in our setting the mode substitute is road rather than rail, the economic mechanism is the same.

7 Welfare

The iceberg trade cost is the farmgate-to-world log price gap (see Section 3):

$$\Delta p_{m,t} \equiv \ln P_t^W - \ln P_{m,t}^D = \ln \tau_{m,t} \tag{25}$$

Because output is predetermined, baseline revenue is

$$\Pi_{m,t}^0 \equiv P_{m,t}^D Q_{m,t}^0 \tag{26}$$

Now consider a wedge change of $g_{m,t}$ log-points, holding $\ln P_t^W$ and $Q_{m,t}^0$ fixed. The domestic price falls by $g_{m,t}$, so new revenue and producer revenue loss are

$$\Pi_{m,t}^{\text{new}} = \Pi_{m,t}^0 \exp(-g_{m,t}) \tag{27}$$

$$L_{m,t} \equiv \Pi_{m,t}^0 - \Pi_{m,t}^{\text{new}} \tag{28}$$

The relative producer revenue loss is therefore

$$\frac{L_{m,t}}{\Pi_{m,t}^0} = 1 - \exp\bigl(-g_{m,t}\bigr) \tag{29}$$

This tradition (Donaldson, 2018; Atkin & Donaldson, 2015; Costinot et al., 2016) motivates Δp as the welfare-relevant sufficient statistic; equation (29) follows from the predetermined-output restriction directly. A positive value lowers revenue on already-harvested output. The fixed quantity assumptions is reasonable considering that planting decisions are committed months before drought realisations, and within-marketing-year supply for storable grains is empirically inelastic (Roberts & Schlenker, 2013).

The pooled second stage estimates a constant elasticity that is a first-order approximation around the baseline (Section 3), and therefore understates the marginal response at the tails of the shortfall distribution, a linearisation limitation revisited in Section 10.3. We therefore first replace $\ln \widehat{X}$ with a shortfall against same-month baseline, since the welfare object is most informative for deviations from typical capacity, and second allow transmission to differ inside documented low-water windows

via episode-specific γ_e, which partially relax the linearity assumption by letting the slope steepen during severe events. The shortfall is

$$S_{m,t} = -\Big(\ln \widehat{X}_{m,t} - \overline{\ln \widehat{X}}_{m,\text{month}(t)}\Big), \tag{30}$$

so $S_{m,t} > 0$ indicates below-baseline capacity. The welfare specification is

$$\begin{aligned} \Delta p_{m,t} = \alpha_m + \lambda_{y(t),m(t)} + \beta\, S_{m,t} \\ + \sum_e \gamma_e \big[S_{m,t} \times \mathbf{1}\{t \in D_e\}\big] \\ + \eta \big[S_{m,t} \times \text{LowAccess}_m\big] + \phi \log(\text{Prod}_{m,t}) + \varepsilon_{m,t} \end{aligned} \tag{31}$$

The episode-specific γ_e treat documented low-water windows as corridor case studies (2016 Tapajós, 2018/2020 Amazon, 2023 Madeira), with $\beta + \gamma_e$ giving the event-specific iceberg passthrough coefficient and η capturing heterogeneity by market access.

The baseline shortfall slope outside documented event windows is small and statistically imprecise ($\widehat{\beta} = 0.028$, $p = 0.31$), indicating that welfare losses idle close to the baseline rather than following a systematic trend. Within documented windows, the transmission coefficient is substantially larger for the two corn-relevant episodes: $\widehat{\gamma}_{2016} = 0.124$ ($p = 0.024$) for Tapajós and $\widehat{\gamma}_{2020} = 0.175$ ($p = 0.004$) for Amazon. The 2018 Amazon estimate is positive but imprecise (0.089, $p = 0.47$); the 2023 Madeira estimate is uninformative for corn (-0.149, $p = 0.27$).

Figure 7 plots the monthly implied losses. The largest losses fall within documented low-water windows: 33.9% in September 2016, 24.7% in July 2020, and 19.7% in October 2020. The 2018 episode peaks at 12.4% in September. The 2023 Madeira result is data-limited. Although the episode is a documented historic drought, the corn panel contains only 10 event observations across 4 municipalities, leaving the estimate uninformative. Figures 8 and 9 therefore use only the 2016 and 2020 corn windows.

Figure 8 shows the distribution of fitted losses by market access conditional on an event shortfall. At the 75th-percentile supported shortfall ($S = 1.46$), the best-access municipality has a fitted loss of 17.6%. The median LowAccess_m municipality has a loss 2.8 percentage-points higher, and the weakest-access municipality 3.2 percentage-points higher. The gradient is steep only at the lower tail of the access distribution and flattens thereafter; the market-access channel is statistically detectable but small relative to the common event shortfall. Figure 9 maps that gradient geographically. The largest increments are in interior Mato Grosso municipalities assigned to Amazon-linked ports.

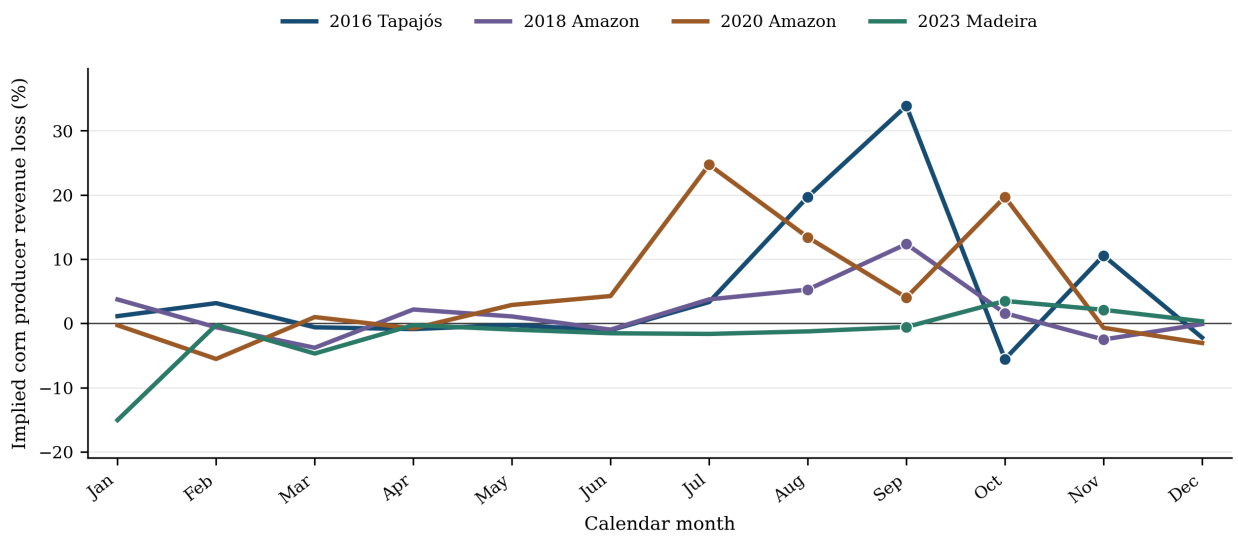


Figure 7: Signed corn incidence across documented low-water episodes. Lines plot monthly mean $100[1 - \exp(-\widehat{g}_{m,t})]$ from equation (31). Filled points mark months inside the documented low-water window. Event-window losses peak at 33.9% in September 2016 and 24.7% in July 2020

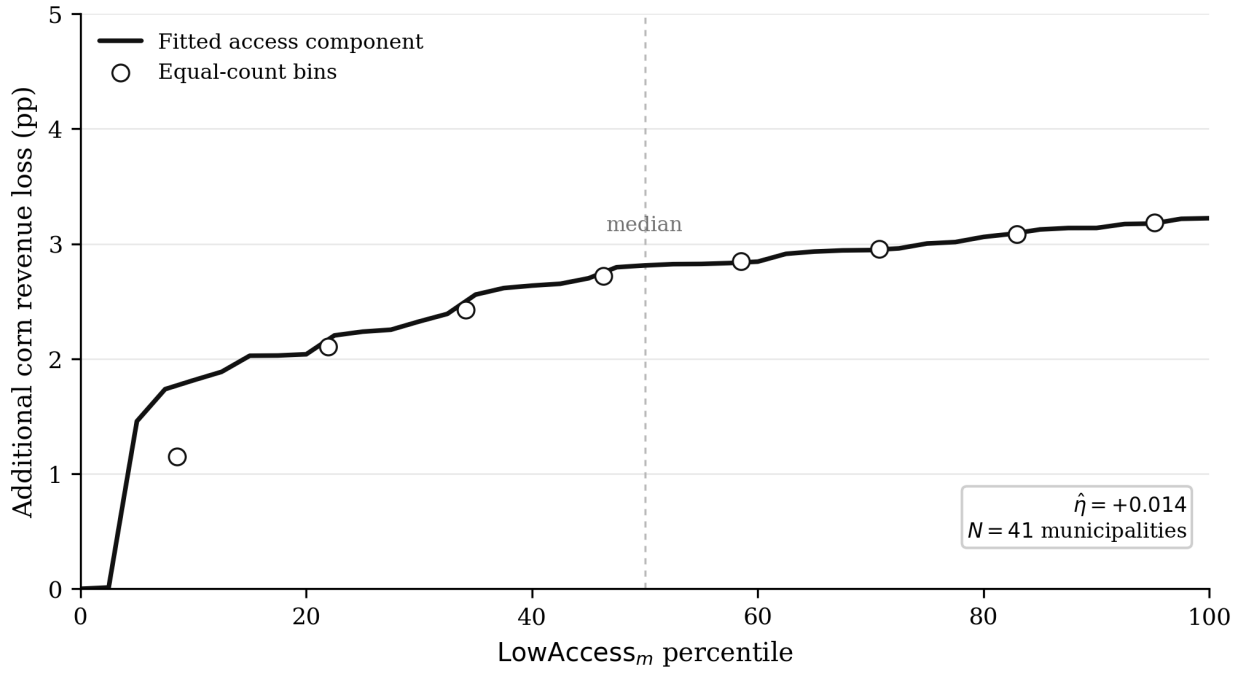


Figure 8: Market access and incremental corn incidence. Municipalities are ranked from 0 to 100 by LowAccess$_m$ within the 41-municipality 2016–2020 event sample; higher percentiles mean weaker fixed access to the assigned export port. The curve reports additional loss, relative to the best-access municipality. The added loss is 2.8 percentage-points at the median and 3.2 percentage-points at the weakest-access municipality.

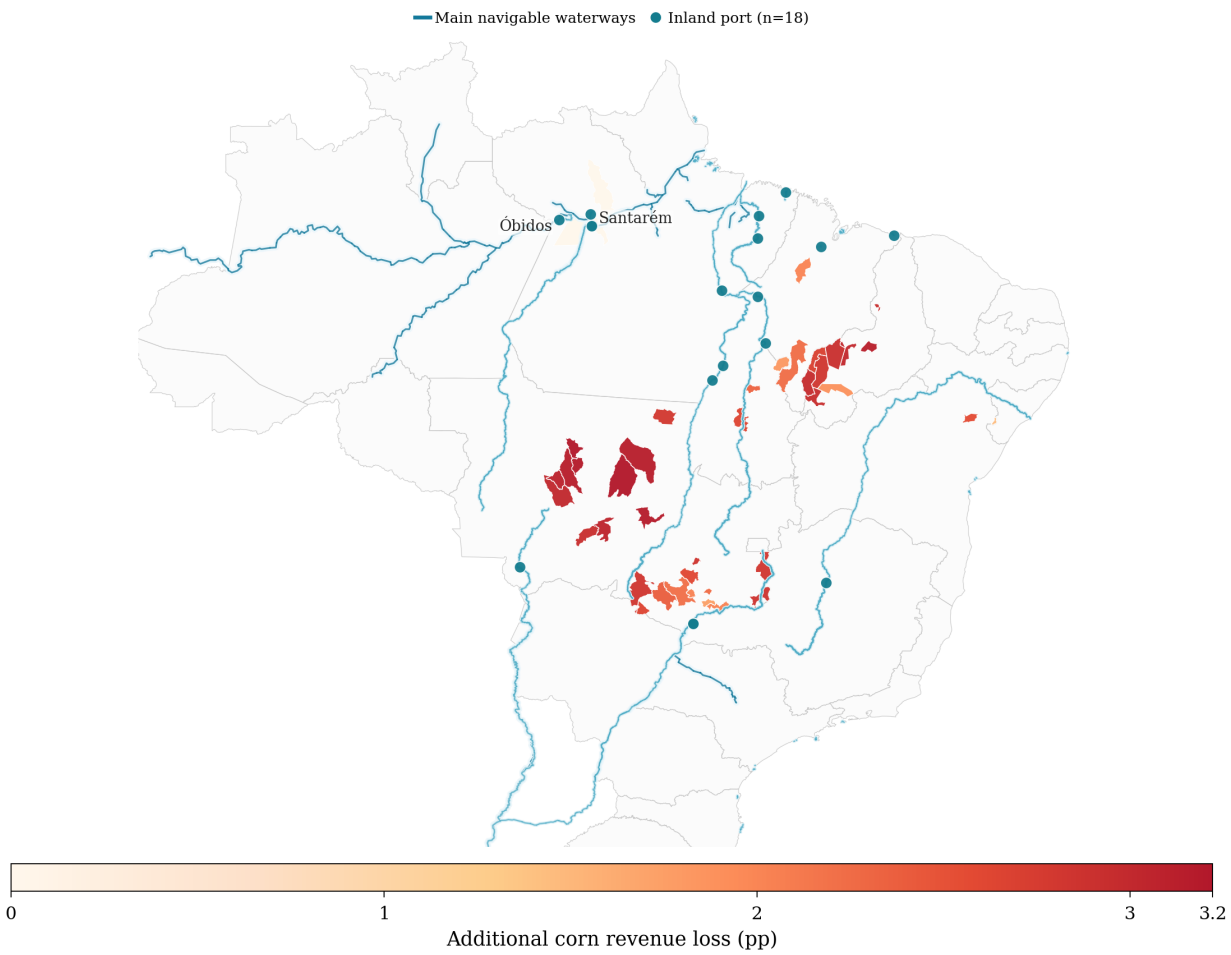


Figure 9: Geography of incremental corn incidence. The map shows the same 41 municipalities and all 13 capacity ports. Municipal colour equals the Figure 8 incremental-loss object; the darkest cluster show the lowest access locations.

8 Discussion

The empirics show that drought-driven river capacity loss widens the inland-to-world price-gap. The welfare analysis translates these elasticities into producer revenue losses. This section interprets what these results imply.

8.1 Episodic spikes of producer losses

The main welfare finding is that producer losses are concentrated in time. While outside observed low-water windows (June-November) the shortfall is near zero, within drought season it rises sharply, drastically lifting losses. This reconciles two magnitudes that otherwise appear inconsistent. Compared to the pooled elasticity from Section 6.3, the welfare analysis shows that conditional on the drought window, revenue impact is an order of magnitude larger. This is because the pooled coefficient is a LATE across all municipalities and months, when the waterway loading constraint

is not binding. However, when the route-minimum depth falls below a threshold, constraint in equation (9) starts binding. The kink around June-July given by $\widehat{\gamma_m}$ in Figure 7 shows this nonlinearity. The sharp contrast in revenue loss between baseline years and severe-drought years reflects the vulnerability of Brazil producers facing climate shock. These drought-window welfare costs are large but in line with external evidence. Steinbach & Zhuang (2025) find the 2022 Mississippi drought cut Louisiana port agricultural export by more than 15 percentage-points during dry-season. While their object is trade-flow volume rather than the producer price wedge, both papers recover the same seasonal concentration of losses around the low-water window.

8.2 Compound climate exposure

Amazonian drought amplifies the producer losses relative to Mississippi because the shock simultaneously hits agricultural production and logistics. The spatial correlation further means a single low-water event propagates across an entire catchment rather than one market. This compounding is spatially uneven. Interior municipalities with weak port connectivity bear larger losses beyond yield loss (Figure 9), so compound climate risk falls disproportionately on areas with more limited market access. This is a reversal of Mitchell & Biram (2025), who find Arkansas markets closest to Mississippi ports experience the largest discounts.

However, the market access is not the primary driver. The welfare estimates attribute most of the event loss to the common capacity shortfall with market access adding only a small increment on top (Section 7). Because that increment is concentrated among the best-connected municipalities (Figure 8), the relevant question is whether a producer has a workable modal alternative at all, not proximity to the port.

8.3 Policy implications

8.3.1 Dredging and maintenance timing

Our first-stage coefficient provides input for evaluating Brazil's dredging programme (Ministério de Portos e Aeroportos, 2026). Given the effect of water-level change on port loading capacity, one can understand how much producer revenue is saved with dredging via the second-stage and welfare analysis. Our results also inform the dredging timing. Figure 7 shows losses concentrate in dry-season months. Pre-dry-season dredging maintenance would help protect both the corn harvest and its shipment window (USDA Foreign Agricultural Service, 2025).

8.3.2 Alternative export routes

Upgrading alternative export routes such as roads and rails moderates the gradient, but Figure 8 shows that the improvement gives limited marginal return. In other words, dredging the existing corridor addresses the larger common shock. Also, this backup option may not hold in the long run. Wang et al. (2024) reveals that road corridors face their own compound exposure, so modal diversification does not fully substitute for waterways.

8.3.3 Risk transfer

Existing Brazilian crop insurance compensates production losses but does not recover revenue loss from transport capacity constraints. A producer who harvests normally but cannot move products to world-market prices is currently unprotected. Index-based insurance products tied to water levels at the binding gauge station may fill this gap. Parametric transport insurance of this kind already exists for Rhine shippers (Marsh, 2024).

9 Robustness

9.1 Placebo test: including non-export products

Besides corn and soybeans, we test sorghum. Sorghum is mostly consumed domestically as animal feed rather than exported through river corridors (Formiga, 2025), so the transport-capacity channel should not apply. The robustness check serves as a placebo test that the throughput instrument affects price gaps only for crops that actually transit through the river network.

Table 9: Second-Stage Results by Commodity: Corn, Sorghum, and Soybeans

	Corn	Sorghum	Soybeans
$\ln(\widehat{X}_{m,t})$	-0.0611^{**}	-0.0271	-0.0465^{***}
	(0.0298)	(0.0785)	(0.0174)
$\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m$	-0.0319^{***}	-0.0365	-0.0271
	(0.0081)	(0.1312)	(0.0238)
$\ln(\text{Prod}_{m,c,t})$	-0.2446^{**}	-0.1878	-0.0269
	(0.1246)	(0.1464)	(0.0612)
Observations	1,822	470	1,257
Municipality FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Municipalities	70	17	45
Clustered SE	Municipality	Municipality	Municipality

Notes: The dependent variable is the log municipality-level crop price gap. Each column estimates the second-stage specification separately by commodity. LowAccess_m is the negative standardised inverse-distance-to-port measure, so higher values indicate weaker fixed market access. Standard errors clustered by municipality are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9 shows that neither coefficients are significant for sorghum inline with our expectation that the instrument moves price gaps for crops that actually transit the river network, though power is limited from fewer observations.

9.2 Mechanism check: the road-diversion margin

The pooled second-stage elasticity bundles several margins from equation (5), congestion, bargaining, and road diversion among them, without identifying any individually. This section tests the road-diversion channel directly.

When droughts constrain barges, goods are substituted onto roads, raising trucking freight rates. If this substitution were frictionless, the diverted goods would be absorbed with minimal impact on trucking freight rates and the price gap outcome. However, Jonkeren et al. (2011) and Mitchell & Biram (2025) both suggest that substitution away from barges is limited and imperfect. Empirically testing whether river-level instrumented barge throughput affects trucking freight rates adds credibility to our second stage by establishing a causal channel and tests imperfect substitutability.

Road freight data comes from SIFRECA (ESALQ-LOG, 2026) and matched on our second stage sample to get 973 route-month observations across 216 routes, 78 months and 17 municipalities concentrated in the Arco Norte agricultural corridor.

We estimate two specifications:

$$\ln(\text{FreightRate}_{r,m,t}) = \alpha_r + \lambda_t + \beta \ln(\widehat{X}_{m,t}) + \phi \ln(\text{Prod}_{m,t}) + \varepsilon_{r,m,t}.$$

$$\ln(\text{FreightRate}_{r,m,t}) = \alpha_r + \lambda_t + \beta \ln(\widehat{X}_{m,t}) + \gamma \left[\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m\right] + \phi \ln(\text{Prod}_{m,t}) + \varepsilon_{r,m,t}$$

where $\text{FreightRate}_{r,m,t}$ is the per-tonne trucking rate on route r originating in municipality m in month t. The rest of the terms follow from our second stage specification. The exclusion restriction for this regression is narrower: minimum water levels must affect trucking rates only through the substitution channel. The most plausible alternative pathway is that drought-induced river-level changes affect hydroelectric generation, which in turn affects diesel prices. However, this would be common to all routes and absorbed by the month fixed effects. Following the theoretical prediction, $\widehat{\beta} < 0$, since a fall in river-level throughput should raise demand for trucking and bid up trucking freight rates.

Table 10: Freight Robustness: Effect of Instrumented Barge Capacity on Trucking Rates

	ln(FreightRate)	
	Basic	Interacted
	(1)	(2)
$\ln(\widehat{X}_{m,t})$	-0.0068^{***}	0.0202
	(0.0025)	(0.0280)
$\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m$		-0.0441
		(0.0465)
$\ln(\text{Prod}_{m,t})$	-0.0024	-0.0051
	(0.0208)	(0.0195)
Observations	973	973
Routes	216	216
Origin municipalities	17	17
Months	78	78
Route FE	Yes	Yes
Month FE	Yes	Yes
Clustered SE	Route	Route

Notes: The dependent variable is the log per-tonne trucking rate on route r. $\widehat{X}_{mt}$ is fitted barge throughput at the Amazon port matched to municipality m from the first-stage regression in Section 5.6. LowAccess_m is the negative standardised inverse-distance-to-port measure, so higher values indicate weaker fixed market access. The sample is matched exactly on municipality, state, and month. Standard errors clustered by route are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10 shows the results of the two estimated specifications. From the base regression, $\widehat{\beta} = -0.0068$, significant at the 1% level, indicating imperfect substitutability in line with previous literature (Jonkeren et al., 2011; Mitchell & Biram, 2025) and supporting the bottleneck mechanism behind our main result. A 30% capacity shock would increase the trucking freight rate by approximately 0.24%, indicative of a large truck market. $\widehat{\gamma} = -0.044$ ($p = 0.34$) is directionally correct but not significant, likely due to a small sample size of 17 municipalities lacking variation in LowAccess_m.

9.3 Exclusion-restriction placebo: the planting window

A potential concern of our estimation is the constraint on agricultural production data. Brazil does not have monthly-level agricultural production by municipality, meaning we use a monthly vegetation index satellite indicator in line with previous literature (Asher & Novosad, 2020; Boudot-Reddy & Butler, 2025). However, the proxy may not capture all producer side shocks threatening the exclusion restriction.

To test this, we construct a placebo from CONAB planting-month calendars (Companhia Nacional de Abastecimento, 2024*a*). During planting the crop is not yet transported, but river levels still matter for yields. A significant placebo coefficient would indicate the instrument is absorbing residual production conditions.

We estimate:

$$\Delta p_{m,c,t} = \alpha_{m,c} + \lambda_{c,t} + \beta \ln(\widehat{X}_{m,t}) + \delta \ln\left(X^{plant,0}_{m,c,s(t)}\right) + \mu \cdot \text{Missing}_{m,c,t} + \gamma\left[\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m\right] + \phi \ln(\text{Prod}_{m,c,t}) + \varepsilon_{m,c,t}.$$

where $\ln\left(X^{plant,0}_{m,c,s(t)}\right)$ averages $\ln(\widehat{X})$ over the planting months of season $s(t)$ and $\text{Missing}_{m,c,t}$ is a dummy absorbing any systematic price-gap difference between the full and subsamples. If insignificant, sample selection is not driving $\widehat{\delta}$. The remaining terms follow Section 6.2. To support the exclusion restriction, $\widehat{\delta}$ should be indistinguishable from zero, $\widehat{\beta}$ should remain comparable to the main result, and $\widehat{\mu}$ should be insignificant.

Table 11: Planting-Window Placebo: Effect of Instrumented Throughput on the Price Gap

	$\Delta p_{m,c,t}$ (1)
$\ln(\widehat{X}_{m,t})$	−0.0193
	(0.0119)
$\ln(\widehat{X}^{plant,0}_{m,c,s(t)})$	−0.0047
	(0.0153)
$\text{Missing}_{m,c,t}$	−0.0896
	(0.1443)
$\ln(\widehat{X}_{m,t}) \times \text{LowAccess}_m$	−0.0194***
	(0.0058)
$\ln(\text{Prod}_{m,c,t})$	−0.2005***
	(0.0719)
Observations	3,079
Municipalities	75
Municipality-commodity entities	115
Months	108
Municipality-commodity FE	Yes
Commodity-month FE	Yes
Clustered SE	Municipality

Notes: The dependent variable is the log municipality-level crop price gap. $\widehat{X}_{m,t}$ is contemporaneous fitted barge throughput from the first-stage regression in Section 5.6. $\widehat{X}^{plant,0}_{m,c,s(t)}$ is the planting-window average of $\ln(\widehat{X})$ for the season $s(t)$ to which observation t belongs, zero-filled where no valid planting observations exist for the entity-season. $\text{Missing}_{m,c,t}$ is an indicator equal to one for entity-seasons without a valid planting placebo. LowAccess_m is the negative standardised inverse-distance-to-port measure. Standard errors clustered by municipality are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The placebo coefficient $\widehat{\delta} = -0.005$ ($p = 0.76$) is indistinguishable from zero, and the missing indicator is null ($\widehat{\mu} = -0.090$, $p = 0.53$), ruling out sample selection bias. $\widehat{\beta} = -0.019$ ($p = 0.11$) is smaller than in Section 6.3 ($\widehat{\beta} = -0.055$) but the interaction $\widehat{\gamma} = -0.019$ ($p < 0.01$) survives at the 1% level. The exclusion restriction is therefore supported by the data.

9.4 Iceberg assumption: world-price pass-through

Our framework treats trade costs as iceberg with no time-varying intermediary mark-ups (Section 3), but Atkin & Donaldson (2015) show this can fail where interior intermediation is concentrated. However, to estimate markups, structural IO literature typically requires quantities, prices and market shares (De Loecker & Warzynski, 2012; Berry et al., 1995). We do not observe market shares or shipping prices and thus cannot formally identify markups. Instead, we test a necessary implication.

Under our framework (Section 3 equation 2), the elasticity of $\ln P^D_{m,c,t}$ to $\ln P^W_{c,t}$ within municipality-commodity should equal one. Incomplete pass-through would indicate variable markups inconsistent with this (Goldberg & Hellerstein, 2008; Burstein & Gopinath, 2014), so we test it with:

$$\ln P^D_{m,c,t} = \alpha_{m,c} + \rho \ln P^W_{c,t} + \varepsilon_{m,c,t}.$$

Table 12: Pass-Through of World Prices to Farmgate Prices

	$\ln P^D_{m,c,t}$
$\ln P^W_{c,t}$	0.946^{***}
	(0.056)
Observations	3,079
Municipality-commodity FE	Yes
Clustered SE	Municipality
Test $H_0 : \rho = 1$	0.929
p-value	0.335

Notes: The dependent variable is the log domestic farmgate price. The coefficient on $\ln P^W_{c,t}$ estimates pass-through from world prices to domestic prices. The specification includes municipality-commodity fixed effects, and standard errors are clustered at the municipality level. The Wald test reports the test of $H_0 : \rho = 1$, corresponding to unit pass-through. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

We test $H_0 : \rho = 1$, the unit pass-through implied by equation (2) and find $\widehat{\rho} = 0.946$ (SE $= 0.056$). The Wald test fails to reject at the 5% level ($p = 0.33$). Under Atkin & Donaldson (2015), unit pass-through is consistent with either perfectly competitive intermediation or with mark-ups that do not respond to world-price movements. Mark-up responses to the capacity shock itself are also not a threat since they are part of the wedge subsumed in $f(\cdot)$, equation (5). The test does not prove perfect competition, but shows the data are consistent with the iceberg framing.

10 Limitations

10.1 Sample selection

The first stage is identified on a heavily filtered subsample of the ANTAQ register. The 1.28 million raw berthing events reduce to 10,342 journeys in the estimation panel. Attrition results from several filters, including restriction to Brazil–Brazil origin-destination pairs, loading-side events, valid IMO numbers, successful waterway routing, and at least one matched gauge within the route buffer. Small, locally operated barges and pusher-tug convoys on the Amazon tributaries are not always

assigned valid IMO numbers, so the estimation sample plausibly over-represents larger, formally registered vessels. If the loading response of these vessels differs systematically from that of the dropped vessels, $\widehat{\pi}$ does not recover the population average, and the second-stage elasticity inherits this measurement error.

10.2 Borderline first-stage strength

The first-stage F-statistic on the excluded instrument is 10.53, which clears the conventional Stock–Yogo threshold of ten for weak instruments under homoskedasticity. The proximate cause of the borderline F is the demanding fixed-effects structure of equation (20). The fixed-effects structure absorbs most of the level variation in both variables, leaving $\widehat{\pi}$ identified from within-vessel and within-route deviations. We view the absence of fixed effects as a more serious issue than the loss of statistical strength. The borderline F is therefore the cost of an identification strategy we believe is more defensible.

10.3 Linearisation and welfare extrapolation

The capacity gradient in equation (7) is a first-order approximation of the trade-cost function $f(\cdot)$ around the baseline, so the estimated $\widehat{\beta}$ and $\widehat{\gamma}$ characterise local properties at average values of $\ln K_{p(m),t}$ and LowAccess_m. The welfare exercise in Section 7 partially relaxes this by allowing episode-specific transmission coefficients γ_e inside documented low-water windows, which let the slope steepen during severe events. The pooled second-stage estimates, however, retain the linear structure. Curvature in $f(\cdot)$ at large shocks could bias the pooled coefficients in either direction. A direct test of curvature using higher-order terms is left to future work.

11 Conclusion

Overall, this paper shows how drought-induced river transport disruptions raises the iceberg trade cost and reduce agricultural market access for inland producers in Brazil. By linking minimum water levels to vessel loading, predicted port throughput, and price gaps, we find river capacity acting as a compounding climate event affecting producers. The first stage confirms this. Minimum water levels significantly predict vessel relative load. The second stage shows that instrumented throughput reductions widen the farmgate-to-world price gap, with a 30% drought-induced capacity fall widening it by 2.4 percentage-points at the median municipality.

The incidence of shocks is spatially uneven. The interaction between fitted throughput and LowAccess_m implies elasticities of 0.067 at the median and 0.072 at the 75th percentile of remoteness. Losses concentrate in dry-season months and are larger where outside transport options are weaker.

These findings reframe river infrastructure as a source of climate shocks challenging the trade buffering mechanism previously identified in the literature. Under drought, corridors become bottlenecks when navigability falls and alternatives are limited. The estimated elasticity chain: minimum water level, vessel load, port throughput to farmgate price provides the empirical inputs for a counterfactual evaluation of Brazil's dredging programme. A dredging policy that raises navigable depth could be compared against programme cost. Policy evaluation should carefully account not only for aggregate throughput gains, but also for protection of remote producers who have larger losses when bottlenecks bind.

Some limitations constrain the scope of these findings. The partial-equilibrium framework abstracts from storage responses, consumer surplus, and long-run route adjustment. Detailed data on firm-level routing and modal freight rates could allow future work to decompose the price-gap elasticity into its constituent margins. Combining the capacity-price relationship with corridor-level climate projections could quantify the future welfare effects of waterways facing recurrent bottleneck risk. As climate change raises the frequency of hydrological extremes across the Mississippi, Rhine, and Mekong, the corridor failure documented here will become a recurring feature of global agricultural supply chains, one that adaptation policy has yet to fully account for.

AI Declaration

We acknowledge the use of Claude Opus 4.7 by Anthropic and ChatGPT 5.5 by OpenAI to assist our preparation for this assessment in line with what was set out as permitted in the assessment guidance. We confirm that no content generated by AI has been presented as our own work.

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